



Controlled formation of BNb_3O_9 nanobelts as superior host material for high performance electrochemical energy storage

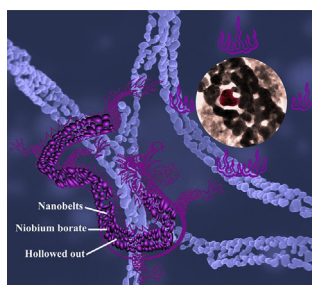
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HIGHLIGHTS

- BNb_3O_9 hollowed-out nanobelts are formed by an adjustable-voltage electrospinning.
- BNb_3O_9 hollowed-out nanobelts show superior performance to the micro-particles.
- $\text{LiFePO}_4/\text{BNb}_3\text{O}_9$ cell presents 66% capacity retention after 200 cycles at 700 mA g^{-1} .
- Structural evolution of BNb_3O_9 upon lithiation is analyzed by in situ XRD.

GRAPHICAL ABSTRACT



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ABSTRACT

Ultra-stable anode materials for lithium-ion batteries are increasing in demand as highly sustainable energy storage system with excellent charge transport becomes unprecedentedly important. Here, a novel Li-ion full cell, utilizing BNb_3O_9 hollowed-out nanobelts as the typical intercalation anode and commercial LiFePO_4 as the cathode in organic electrolyte, is reported. Moreover, the hollowed-out framework of BNb_3O_9 nanobelts, prepared by an adjustable-voltage electrospinning method, exhibits prominent cyclic stability (116.1 mAh g^{-1} at 700 mA g^{-1} (12C) over 1000 cycles) and outstanding rate capability (126.8 mAh g^{-1} at 900 mA g^{-1}) in half cell. The key finding is that the formation mechanism of BNb_3O_9 hollowed-out nanobelts can be thoroughly discussed by scanning electron microscope. The contribution of the external pseudocapacitance largely affects the rate capability, which is determined via kinetic analysis. Furthermore, the lithium storage mechanism and structural evolution of BNb_3O_9 are investigated through in-situ X-ray diffraction. Such preeminent electrochemical performance and distinctive morphology endow BNb_3O_9 hollowed-out nanobelts with potentials as anode materials for lithium-ion batteries.

1. Introduction

Rechargeable lithium-ion batteries (LIBs) are extensively considered as one of the most advanced and vital energy storage devices owing to their high power density and low impact to environment [1–7].

Graphite as a commercialized anode material is successful for LIBs because of its low cost and relatively large theoretical capacity (372 mAh g^{-1}). However, lithium intercalation at low operating voltage (about 0.1–0.2 V vs. Li^+/Li) may lead to the formation of lithium dendrites during high-rate charge and discharge [8–10]. Therefore, it is

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necessary to explore alternative anode materials which are able to react with lithium at voltages above 0.5 V in the organic electrolytes.

In this aspect, titanium-based oxides have attracted widespread attention because of their suitable lithium ion insertion/extraction voltage window of 1.0–2.0 V. Particularly, the zero-strain $\text{Li}_4\text{Ti}_5\text{O}_{12}$ with spinel structure is reported to be a dominant candidate for high-performance LIBs [11–17]. However, its low theoretical capacity cannot satisfy the demand of current market, which is the main drawback of $\text{Li}_4\text{Ti}_5\text{O}_{12}$ [18,19]. Additionally, the other titanium-based oxides such as TiO_2 , LiCrTiO_4 and $\text{LiTi}_2(\text{PO}_4)_3$, also show the low theoretical capacity ($< 200 \text{ mAh g}^{-1}$) with a $\text{Ti}^{4+}/\text{Ti}^{3+}$ redox couple [20–25]. Recent research trends in exploring anode materials for LIBs have been devoted to niobium-based oxides because of their relatively high theoretical capacity (389 mAh g^{-1} , supposing $2e$ transfer formula unit). Besides, $\text{Nb}^{5+}/\text{Nb}^{4+}$ and $\text{Nb}^{4+}/\text{Nb}^{3+}$, as two pairs of redox couples, are located within the voltage range of 1–3 V, which may avoid the formation of the solid electrolyte interface (SEI) film during charge and discharge cycles [26–31].

Herein, we report a facile and efficient approach to synthesize BNb_3O_9 hollowed-out nanobelts (denoted as BNO-H) by electrostatic spinning technology for the first time. The resulting hollowed-out banded structure is characterized by interconnected nanocrystallines with an average diameter of 100 nm and macropores separated by these nanocrystallines. Excitingly, the BNO-H shows enhanced electrochemical performances. Moreover, we find that in the voltage window of 1.0–3.0 V, five lithium ions can be stored in BNO-H structure per formula unit during discharge process, leading to a discharge capacity of 309 mAh g^{-1} . In order to further insight into the lithium storage mechanism, we investigated the structural change of the BNO-H electrodes based on in-situ X-ray diffraction technique. The results display that BNO-H suffers a complicated two-phase reaction and three solid solution processes. The BNO-H shows not only ultrahigh reversibility, but also a high capacity, indicating it is a pretty hopeful anode material for LIBs.

2. Experimental

2.1. Synthesis of BNb_3O_9 nanomaterials

The raw materials of poly (vinylpyrrolidone) (PVP) (MW ≈ 1300000 , Macklin), H_3BO_3 (AR, Macklin) and $\text{C}_{10}\text{H}_5\text{NbO}_{20}$ (AR, Macklin) were commercially available. The distilled water and absolute ethanol at a volume ratio of 1:4 were used as the solvent. After that, 1.116 g of $\text{C}_{10}\text{H}_5\text{NbO}_{20}$ and 0.043 g of H_3BO_3 were added into the above solvent to form a mixed solution. Then, 1.6 g PVP was dissolved in the above mixed solution (16 mL). After magnetic stirring for 12 h, the obtained precursor solution was applied to electrostatic spinning. Then the precursor solution was transformed into a syringe with a stainless steel nozzle (21-gauge) as the anode. The plastic syringe was placed vertically and the typical fixed distance between the top of the nozzle and the grounded collector acting as the cathode is 18 cm. A voltage of 16 kV was applied between the two electrodes through a high voltage power supply. The precursor fibers collected on the collector were dried in a vacuum drying oven at 80°C . Subsequently, the dried precursor fibers were calcined to 750°C for 8 h in air atmosphere to obtain target samples. To fully understand the formation mechanism of BNb_3O_9 hollowed-out nanobelts (denoted as BNO-H) and its suitability for the application, a series of the varying applied voltages (V) were investigated. The corresponding precursor fibers were designated as BNO-V (V = 8, 13 and 16 kV), as shown in Table S1.

2.2. Synthesis of BNb_3O_9 microparticles

BNb_3O_9 microparticles (BNO-M) were synthesized by high temperature solid state reaction. First, 0.538 g of H_3BO_4 (AR, Macklin) and 2.759 g of Nb_2O_5 (AR, Macklin) were mixed by a high-energy ball-

milling machine for 12 h in ethanol medium. After ball milling, the slurry was dried at 80°C for 24 h. Then, the obtained mixture was heated to 950°C for 12 h in air atmosphere.

2.3. Characterizations

Powder X-ray diffraction (XRD) was characterized using a Bruker AXS D8 Focus diffractometer (Cu $K\alpha$ radiation). In-situ XRD measurements were analyzed by using the aforementioned XRD equipment. The morphology and nanostructure of samples can be observed by the scanning electron microscope (SEM, Hitachi SU-70) and transmission electron microscopy (TEM, JEOL JEM-2010). Elemental analysis was conducted by high-resolution energy transmission electron microscopy (HRTEM, JEOL JEM-2010) rigged with an energy dispersive X-ray spectrometer (EDS). The elemental valence states in BNO-H were characterized by X-ray photoelectron spectroscopy (XPS, Theta Probe AR-XPS System, Al $K\alpha$ radiation).

2.4. Electrochemical measurements

For half cell tests, CR-2032 coin cells were assembled in an argon-filled glove box. The working electrodes were composed of active materials, acetylene black, and poly (vinylidene difluoride) binder at a weight ratio of 8:1:1 into N-methyl pyrrolidone. The electrodes were coated on a copper foil and dried at 100°C for 12 h in vacuum. The mass loading of active materials was $\sim 2 \text{ mg cm}^{-2}$. The coin-type half cells consisted of the BNO-H as the working electrode, lithium metal as the counter electrode, 1 mol L^{-1} LiPF_6 in ethylene carbonate/dimethyl carbonate as the electrolyte, and Whatman glass fiber as the separator. For full cell tests, the BNO-H anode was coupled with a commercialized LiFePO_4 cathode. The mass loadings of the BNO-H and LiFePO_4 were in the range of 1.0–2.0 mg cm^{-2} . The cell electrochemical performance was examined using a multichannel LANHE CT2001A battery testing system instrument at room temperature. Then, cyclic voltammetry (CV) was carried out using a Bio-Logic VSP electrochemical workstation from 1.0 to 3.0 V. In addition, electrochemical impedance spectroscopy (EIS) analyses were carried out on a CHI660D electrochemical workstation with a frequency range from 0.01 to 100000 Hz.

3. Results and discussion

The general formation procedure of complex one-dimensional (1D) nanostructures is studied via electrospinning. Firstly, the cross-section of the initial liquid jet is often in circular shape due to the applied static voltage. Secondly, the ejected circular precursor undergoes a transient movement towards the grounded collector driven by electrostatic field force. Thirdly, the spun precursor fibers are solidified and collected over the grounded collector. Based on the mentioned analysis, the electrospinning schematic illustration of precursor fibers formed at 8, 13 and 16 kV applied voltages (denoted as BNO-8, BNO-13 and BNO-16) is shown in Fig. 1. We further propose a transformation mechanism for the formation of the BNO-16, which mainly includes three reasons. As the previous works reported, during the process of moving the precursor fibers to the collector, a dry surface of precursor fibers is formed due to the evaporation of the ethanol at 40°C [32–34]. Moreover, the various speeds of movement caused by different applied voltages can lead to the different evaporation times of the ethanol, which eventually leads to the formation of the dry surfaces with different thicknesses around the outer layer of the precursor fibers. When a relatively low applied voltage (8 kV) is fixed on the anode (Figure S1), the BNO-8 will spend more time flying from the anode to the ground collector by reason of the small electrostatic force [35], resulting in the formation of a thicker dry surface. Accordingly, once a sufficiently high voltage (16 kV) is applied, the BNO-16 will expend less time due to the larger electrostatic force, forming a thin dry surface in the end. Subsequently, the continuous evaporation of the internal ethanol will cause

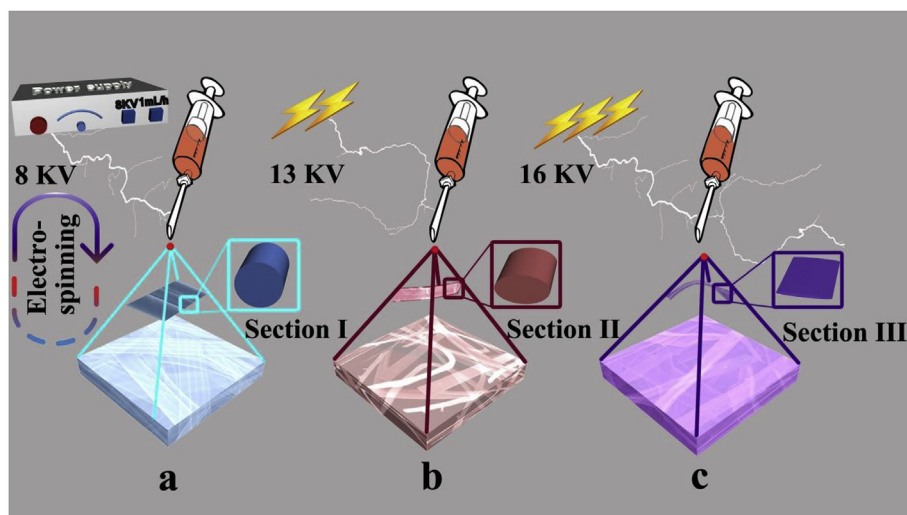


Fig. 1. Schematic illustration of the fabrication strategy for BNb_3O_9 precursor fibers with different cross sections.

the collapse of the BNO-16. It depends on the thickness of the dry surfaces that is formed, because different thicknesses of the surfaces have a direct relation with the anti-collapse capabilities. Once the spinning voltages are fixed at 13 kV, the BNO-13 with the elliptical cross section can be gained (Figure S2). Similarly, the BNO-16 with a more flat elliptical cross section could be gained when the working voltage increased to 16 kV (Figure S3). Consequently, the results of these experiments strongly suggest that the applied voltages play a decisive role in the formation of the BNb_3O_9 hollowed-out nanobelts (denoted as BNO-H), allowing the cross section shapes of the precursor fibers to be controlled.

Fig. 2a, d, and g illustrate the schematic models of the calcined

samples with corresponding tailored applied electrospinning voltages of 8, 13 and 16 kV. As might have been expected, the morphology evaluation of the calcined samples with different applied voltages are from typically nanowires (denoted as BNO-8-W) to nanobelts (denoted as BNO-13-B), and finally to BNO-H. The average diameters of the BNO-8-W, BNO-13-B, and BNO-H are 1 μm , 1.4 μm , and 550 nm, respectively. When a relatively low applied voltage (8 kV) is fixed on the anode, the corresponding BNO-8-W has a typical circular cross section (Fig. 2b, c, and S4a, S4b). Similarly, when the applied voltage increases to 13 kV, the cross section morphology of the BNO-13-B becomes representatively elliptical (Fig. 2e, f, and S5a, S5b). Once the applied voltage rises further to 16 kV, the BNO-H appears to be hollowed-out nanobelts with

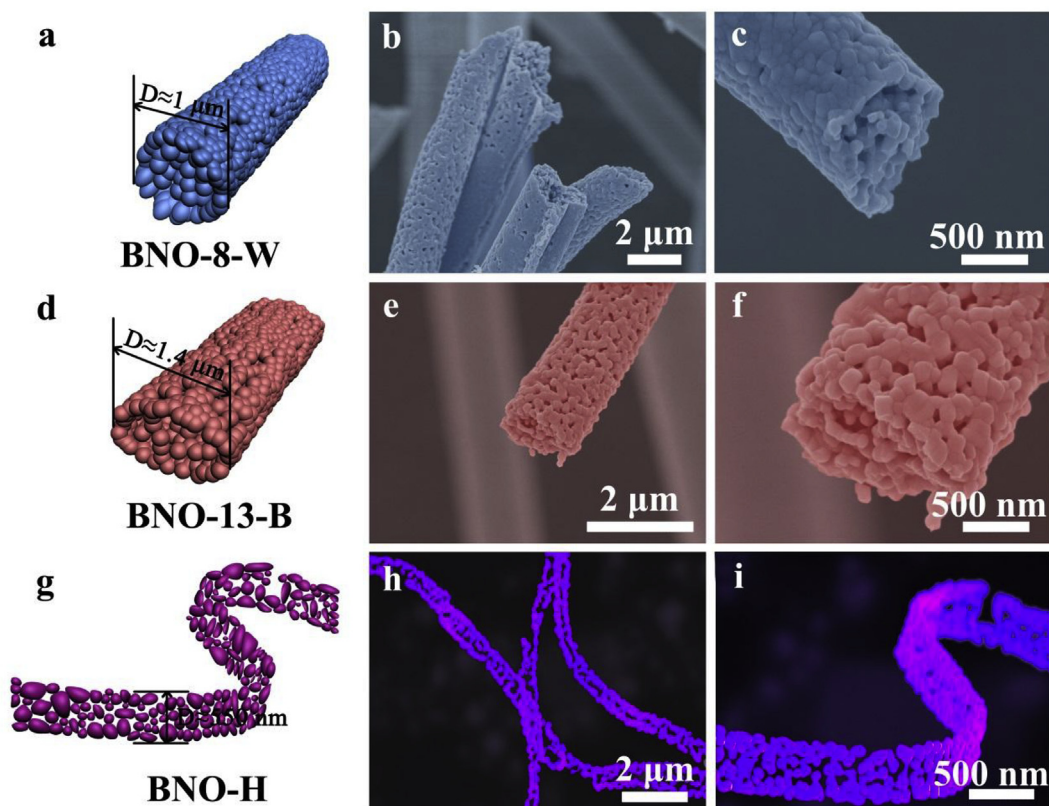


Fig. 2. (a) Schematic illustration of the structural features of the BNO-8-W; (b, c) SEM images of the BNO-8-W; (d) Schematic illustration of the structural features of the BNO-13-B; (e, f) SEM images of the BNO-13-B; (g) Schematic illustration of the structural features of the BNO-H; (h, i) SEM images of the BNO-H.

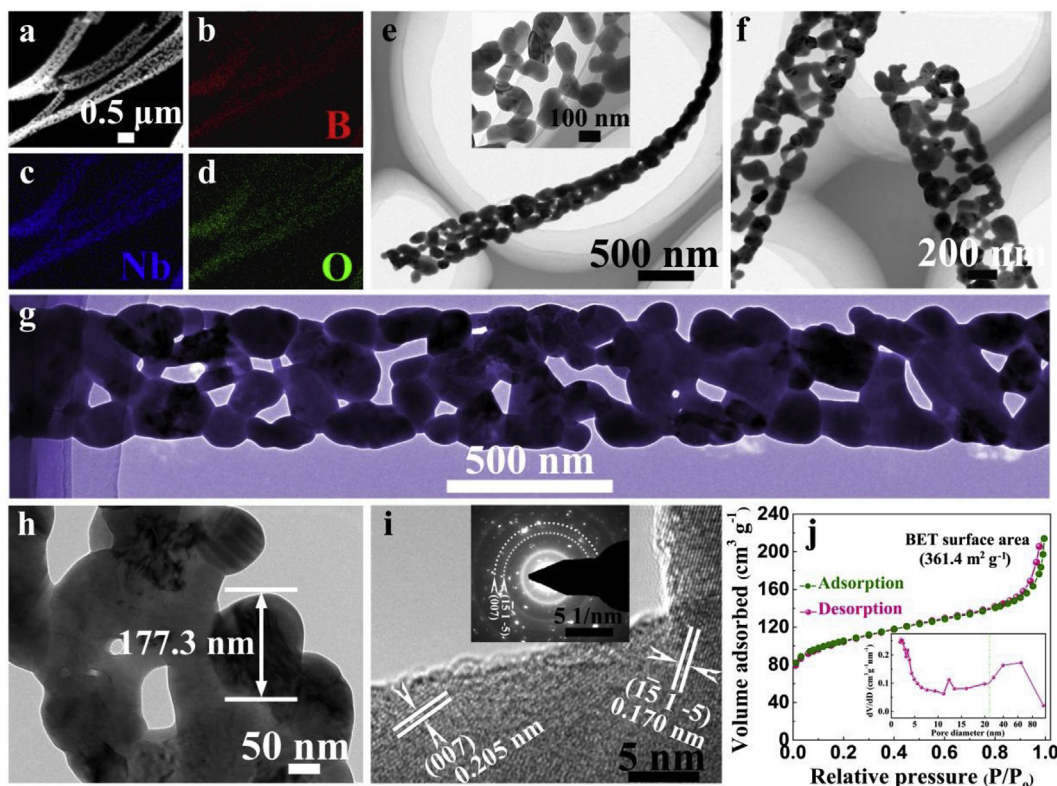


Fig. 3. (a) SEM image, (b–d) EDS elemental mappings of the BNO-H; (e–h) TEM images, (i) HRTEM image of the BNO-H; (j) Nitrogen adsorption isotherm curve and BJH adsorption pore size distribution plot (inset) of the BNO-H.

typical rectangular cross sections (Fig. 2h and i). Further XRD analyses of the BNO-8-W, BNO-13-B, and BNO-H disclose that they are also pure tetragonal BNb_3O_9 (Figures S4c, S5c, and S6a). Moreover, to verify the valence states of BNb_3O_9 , the BNO-H is also examined by XPS. The results are shown in Figures S6b–6d. The XPS spectrum of B 1s single peak is presented in Figure S6b and displays energy at about 190.3 eV, corresponding to B 1s. The two peaks at 210.1 eV and 207.4 eV are assigned to Nb 3d 5/2, revealing the Nb (V) oxidation state (+5) of BNb_3O_9 (Figure S6c) [36]. The main peak of O 1s (Figure S6d) at 529.6 eV is ascribed to the Nb–O bands. Another weaker peak at 531.1 eV corresponds to the oxygen in hydroxyl group [37].

The EDS elemental mapping of the BNO-H not only shows a considerable amount of hollowed-out nanobelts but also confirms the uniform distribution of B, Nb and O elements on the nanobelts (Fig. 3a–d). Correspondingly, the specific compositions of the BNO-H are determined by EDS spectrum supplied in Table S2. Besides, the other EDS surface scan across BNO-H further proves the presence of hollow space and three elements in nanobelts (Figure S7). The morphology of the BNO-H is characterized by SEM and TEM (Fig. 3e–h and S8). The BNO-H shows one-dimensional banded morphology with a length up to several microns (Fig. 3e). The inset in Fig. 3e provides the magnified HRTEM image of the BNO-H, from which a width of 500 nm can be discerned. The hollowed-out banded structure and aperture size can be observed in Fig. 3f and g. This structure facilitates the effective transport of lithium ions. The single grain size of the BNO-H is 177.3 nm as shown in Fig. 3h. HRTEM image in Fig. 3i shows that the lattice fringes of 0.205 nm and 0.170 nm can be indexed to the (007) and (15 $\bar{1}$ –5) planes of BNb_3O_9 (JCPDS 19-0867). Selected area electron diffraction (SAED) pattern also indicates the polycrystalline characteristics of the BNO-H (inset in Fig. 3i). Apparently, these evidences suggest the purity and the polymorphism of the BNO-H. Besides, a N_2 adsorption isotherm experiment is conducted for the BNO-H. The specific Brunauer-Emmett-Teller (BET) surface area of the BNO-H is

$361.4 \text{ m}^2 \text{ g}^{-1}$ (Fig. 3j). Moreover, the surface areas of the BNO-8-W ($72.4 \text{ m}^2 \text{ g}^{-1}$) and the BNO-13-B ($324 \text{ m}^2 \text{ g}^{-1}$) are smaller than that of the BNO-H (Figures S4d and S5d). The strong N_2 adsorption pressure is lower than the relative pressure of 0.1, which means the existence of micropores. The increase of the N_2 adsorption isotherms ranging from 0.1 to 1 is characteristic of mesopores (0.1–0.6) and macropores (0.6–1) [38]. Pore-size distribution curve in the inset of Fig. 3j shows BNO-H with mesopores peaking at 3.5 and 12.3 nm, and macropores peaking at 64.3 nm. These results are in good agreement with the TEM observations of a reticular porous structure (Fig. 3e–h). From the above results, the BNO-H with high specific surface area are beneficial to contact with the electrolyte and promote lithium ions transport especially at high rates.

Then the BNO-H is evaluated as an anode material for LIBs, demonstrating its promising application. Fig. 4a exhibits the emblematical galvanostatic charge-discharge curves of the BNO-H in the voltage range of 1–3 V at the a current density of 100 mA g^{-1} . The first discharge and charge specific capacities are 294.2 and 245.1 mAh g^{-1} , respectively, which are much higher than that of the BNO-M (Figure S9a). The irreversible capacity loss in the first cycle is 49.1 mAh g^{-1} , which is mainly caused by the formation of the SEI film and the decomposition reaction of the organic electrolyte [39–41]. Moreover, the discharge and charge plateaus of the BNO-H are located at 1.57 and 1.76 V, respectively. As shown in Fig. 4b, a pair of sharp reduction/oxidation peaks of the BNO-H is corresponding to discharge and charge plateaus in the cyclic voltammetry plots. There is good overlap between the second scan and third scan, which means the great cyclability of the BNO-H. Furthermore, for comparison, the difference in polarization can be delivered from Figure S9b. It shows that BNO-H has a lower polarization (0.19 V) than BNO-M (0.23 V). And the value of redox polarization for BNO-H is also smaller than that of Nb_2O_5 microsphere (0.35 V) [42] as well as spinel LiCrTiO_4 particle (0.4 V) [43]. Hence, the result is that BNO-H possesses a fast Li^+ insertion/extraction process

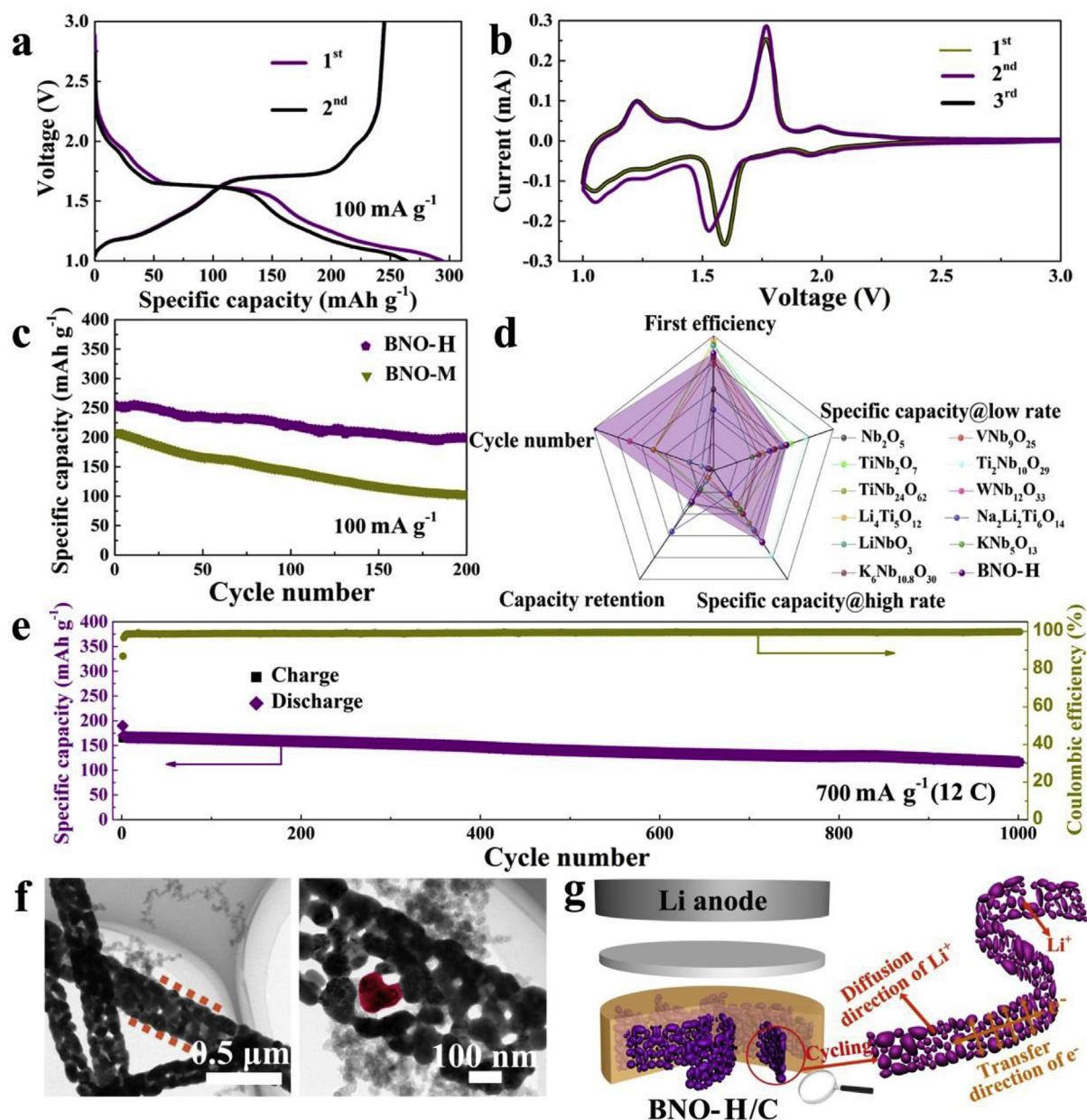


Fig. 4. (a) Discharge/charge voltage profiles at 100 mA g^{-1} , (b) CV curves, (c) cycling performance at 100 mA g^{-1} , (d) comparison of the electrochemical properties, (e) cycling performance at 700 mA g^{-1} ; (f) TEM images of the BNO-H after 1000 cycles; (g) three-dimensional view (left) and simplified two-dimensional plane view (right) of the BNO-H electrode.

and faster reaction kinetics. It is expected that BNO-H can be an alternative for a high-rate anode.

The BNO-H exhibits excellent cycle performance and rate capability as shown in Fig. 4c, S9c and S9d. At the current density of 100 mA g^{-1} , BNO-H achieves a high reversible specific capacity of 198.5 mAh g^{-1} at the 200th cycle, which is 80% of the initial charge capacity. In comparison, the BNO-M suffers from rapid capacity decay since it only retains a capacity of 104 mAh g^{-1} at the 200th cycle (Fig. 4c), indicating a relatively poor property of bulk-material as the anode electrode. When cycled at 500 mA g^{-1} (Figure S9d), the BNO-H still maintains a relatively high initial charge capacity of 199.4 mAh g^{-1} , retaining 82.2% of the initial capacity after 400 cycles. The average coulombic efficiency over 400 cycles is calculated to be 99%, which also demonstrates the superiority of the BNO-H. In addition, the surpassing rate performance of the BNO-H at different current densities from 100 to 900 mA g^{-1} is proposed (Figure S9c). It delivers large capacities of 246.6, 240.6,

232.3, 215.9, 199.3, 183.8, 165.2, 146.1 and 126.8 mAh g^{-1} , respectively. On the contrary, the reversible capacity of the BNO-M rapidly decreases from 202.5 to 67.6 mAh g^{-1} (Figure S9e). Besides, the reversible capacities of BNO-M severely decrease from 155.8 to 95.3 mAh g^{-1} after 400 cycles at 500 mA g^{-1} (Figure S9f). Obviously, benefiting from the special morphology, BNO-H electrode with excellent rate property is realized. As a reference, we design a radar map (Fig. 4d) to compare our work with those results in recent literatures (see detailed information in Table S3) [44–54]. It can be clearly seen that the size of the purple shaded area is the largest in the radar map. The capacity performance and rate property of BNO-H are at the upper level among the typical lithium ion battery anode materials. This means that the hollowed-out banded structure is beneficial to optimize the electrochemical performance of BNb_3O_9 . Finally, BNO-H also shows a notable pseudocapacitive effect, contributing to the extra specific capacity (Figure S10). Owing to their synergistic effect, the rapid transfer of

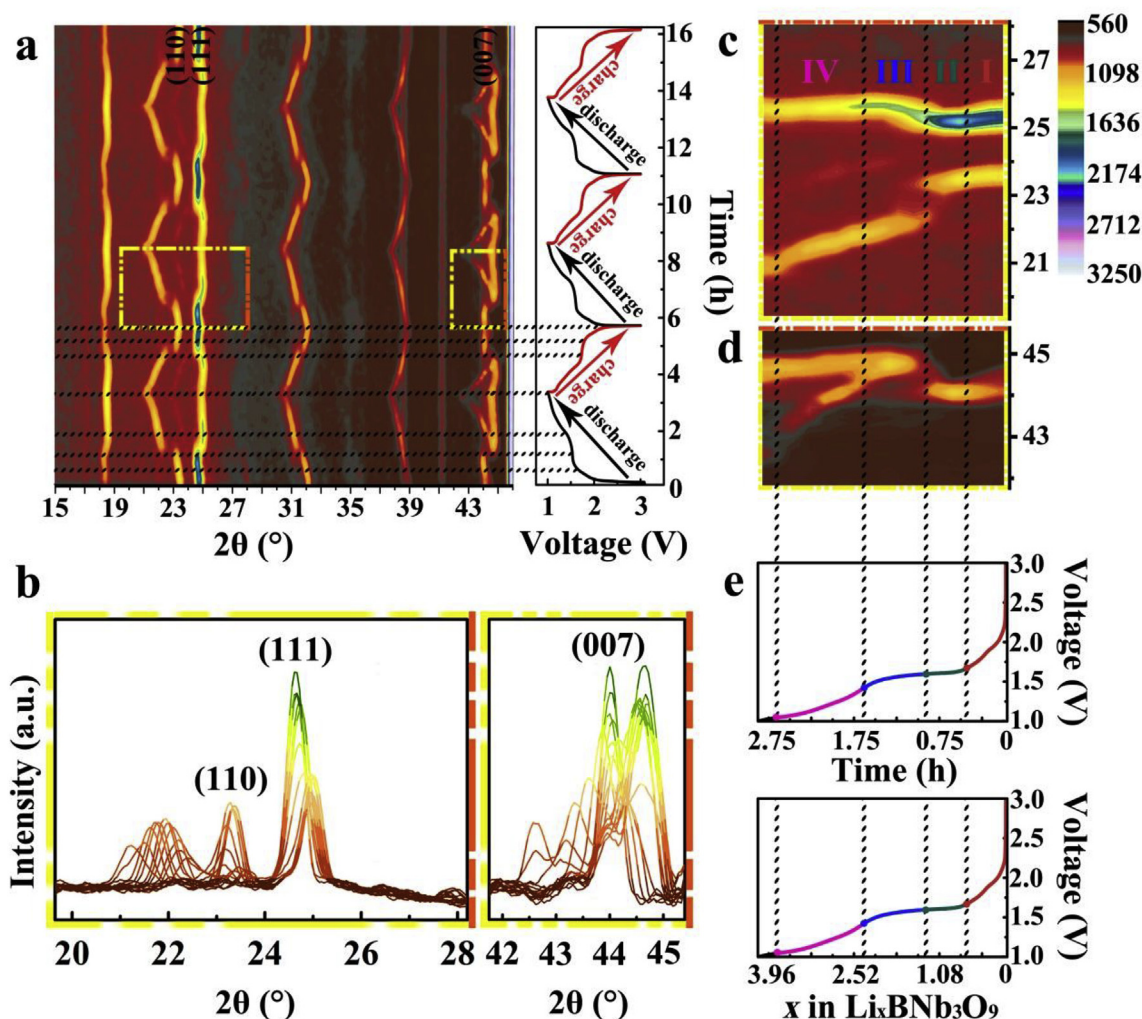


Fig. 5. (a) In-situ X-ray diffraction patterns selected from 15° to 46°, (b) Single-peak spectrums, (c) 19.6° to 28.2°, (d) 41.7° to 45.7° of the BNO-H. (e) The corresponding voltage-composition profile and discharge curve of the BNO-H.

electron/lithium ions and the large degree of the electrochemical reaction can be easily achieved, which respectively lead to the excellent rate capability and the large specific capacity of BNO-H.

The long-term cycling stability at a high current density of 700 mA g^{-1} (12C) is also evaluated. As Fig. 4e indicates, a significant reversible capacity of 116.1 mAh g^{-1} can be maintained even after 1000 cycles, which is 71% of the initial charge capacity. The capacity retention of BNO-H is better than that of BNO-M (capacity retention is 54.3%) under the same experimental condition (Figure S9g). The cycling Coulombic efficiency at high current density for the BNO-H electrode is nearly 100% after the first several cycles. To examine the structural stability of the materials, we carry out the postmortem studies by HRTEM examinations after 1000 cycles at a high current density of 700 mA g^{-1} (Fig. 4f). Clearly, the size, shape and nanoparticle integrity of the BNO-H are still well retained. Meanwhile, the BNO-M clumps suffer from massive structural destruction after cycling (Figure S11). Electrochemical impedance spectroscopy (Figure S12) analysis shows that BNO-H electrode manifests the low charge transfer resistance during lithiation/delithiation [55]. Moreover, the resistance of the prepared BNO-H electrodes is lower than that of other niobate anode electrodes (Table S4) [45–49,51].

As previously discussed, 1D hollowed-out nanobelts produce a synergistic effect during lithiation/delithiation (Fig. 4g) and can improve the overall electrochemical performance. To clarify this synergistic effect of BNO-H, in-situ XRD measurements are executed to monitor the

phase transformations accompanying the charge and discharge of the Li/BNO-H cell. In the Li/BNO-H cell, the diffraction pattern is obtained in the 2θ range of $15.0\text{--}46.0^\circ$, as shown in Fig. 5a, the periodic shifting of the (110), (111) and (007) reflection peaks suggests the occurrence of reversible phase transformations during lithium ion intercalation/extraction. The corresponding entire diffraction pattern is provided to confirm the quasi-reversible of the structure (Figure S13). The (110) and (007) characteristic peaks shift toward lower angles when the lithium ions are inserted into the BNO-H. Conversely, these two peaks return to their original positions in the charge process, corresponding to the conventional routines of lithium storage. Moreover, the whole XRD patterns of BNO-H exhibit distinct reproducibility during repeated lithiation/delithiation. In addition, the (111) reflection peak position of BNO-H fluctuates regularly according to the cycling, and it shifts slightly. These first two characteristics reveal that the BNO-H has excellent structural reversibility during the lithiation/delithiation.

To further demonstrate above mechanisms, we elucidate the time-resolved phase evolution for the second discharge process of the BNO-H by combining in-situ XRD intensity plots, voltage-composition profile and discharge curve (Fig. 5b–e). During the discharge process, four stages of phase evolutions are observed. Based on the diffraction peak changes, it is indicated that the lithium ion intercalation process of BNO-H contains a complicated two-phase reaction and solid solution. In the early stages of lithium ion intercalation (I region, the second discharge process from 0 to 0.43 h), the initial BNb_3O_9 phase transforms to

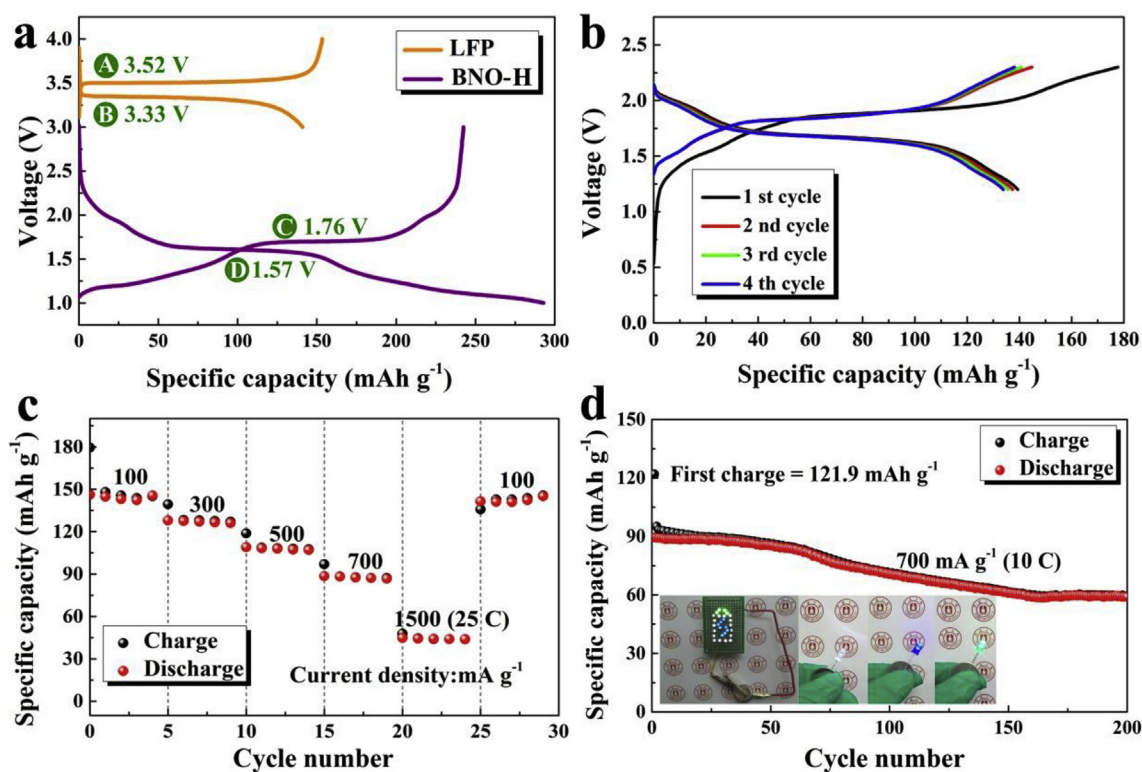


Fig. 6. (a) Charge/discharge voltage profiles of LiFePO₄ and BNO-H half-cells at a current density of 100 mA g⁻¹; (b) Charge/discharge curves of LiFePO₄/BNO-H full-cell at a current density of 100 mA g⁻¹; (c) Rate performance of LiFePO₄/BNO-H full-cell at a current density range of 100–1500 mA g⁻¹; (d) Long-cycle performance of LiFePO₄/BNO-H full-cell at a current density of 700 mA g⁻¹ with an inset of light emitting diode array powered by this full cell.

Li_{0.67}BnB₃O₉ with a gradually decrease of the peak intensity, which is corresponding to a solid solution reaction. The II region is also a solid solution domain (the sequential discharge process from 0.43 to 0.91 h), in which the Li_{0.67}BnB₃O₉ phase transforms to Li_{1.34}BnB₃O₉, corresponding to no variation of the crystal system, but resulting in further intensity attenuation and low angle offset (Figures S14a, S14c). When the discharge time reaches 1.64 h (III region), the Li_{1.34}BnB₃O₉ phase diffraction peaks gradually vanish along with generation of a new Li_{2.43}BnB₃O₉ phase, indicating an apparent two-phase reaction zone. Furthermore, a flat plateau of the discharge curve corresponds to this two-phase reaction zone. At the end of the two-phase zone, the IV region is observed which corresponds to a large solid solution in the curve. As seen in Figures S14b and S14d, these reaction processes are highly reversible as the initial diffraction patterns are recovered completely at the end of the charging.

The excellent cycling property, good rate capability, and high capacitance of the BNO-H half-cell encourage us to further explore its practical application potential for full-cell system. LiFePO₄ is a widely accepted cathode for high-power LIBs. As reported in Fig. 6a, the orange charge/discharge curves for the commercialized LiFePO₄ show two plateaus at 3.52 and 3.33 V, corresponding to Fe²⁺/Fe³⁺ redox reactions. And the specific discharge capacity of the LiFePO₄ cathode at a current density of 100 mA g⁻¹ is 141 mAh g⁻¹ in the first cycle. By using this LiFePO₄ (72% excess) as cathode and the BNO-H as anode, a practical full cell is constructed. Fig. 6b shows the charge/discharge curves and the practical specific capacities of the LiFePO₄/BNO-H full cell. At a current density of 100 mA g⁻¹, the full cell shows an initial charge capacity of 177.7 mAh g⁻¹ and a discharge capacity of 139.4 mAh g⁻¹, corresponding to an initial Coulombic efficiency of 78.4%, which can be assigned to the formation of lithium ion SEI layers and irreversible reactions on both electrodes. Upon repeated cycles, the full cell shows the Coulombic efficiency approaching 100% (except the first six cycles for the activation) and low capacity fading over 50 cycles

(Figure S15). Remarkably, the rate performance of the LiFePO₄/BNO-H full cell with a current density range of 100–1500 mA g⁻¹ is given in Fig. 6c. The specific capacity of the full cell remains 89 mAh g⁻¹ at a current density of 700 mA g⁻¹. Even when the current density is further increased to an ultrahigh value of 1500 mA g⁻¹ (25C), the capacity of full cell still reaches 47 mAh g⁻¹. Fig. 6d gives the long-life electrochemical storage capability of LiFePO₄/BNO-H system at 700 mA g⁻¹, which presents stable cycle performance with 66% capacity retention after 200 cycles. Thus, it can be the power source for the light-emitting diodes as illustrated schematically in the inset of Fig. 6d. Such full lithium-ion batteries endow their potential for practical applications as large-scale energy storage devices.

4. Conclusions

The designed concept for synthesizing the BNO-H presents in this work is easily reproducible and facile. The growth of 1D hollowed-out banded structure is controlled by a simple electrospinning technique with different applied voltages. This provides an alternative approach for fabricating valuable hollowed-out banded materials by varying applied voltages.

By right of the distinctive hollowed-out banded structure, the BNO-H shows ascendant electrochemical performance as the anode for lithium ion batteries. The superior performance of BNO-H can be related to the following aspects. First, the nanoparticles of the BNO-H cling tightly to maintain the structural integrity. Second, the 1D hollowed-out banded structure enables a short diffusion distance for fast lithium ion diffusion during lithiation/delithiation, and a sufficiently large electrode/electrolyte interface for the rapid charge transfer reaction. Third, the pores and the free space secured by nanobelts effectively alleviate the volume changes during lithium insertion and extraction.

In summary, benefiting from the advantageous architecture, the BNO-H exhibits remarkable electrochemical performance as anode

materials for lithium ion batteries with a high specific capacity up to 245.1 mAh g^{-1} at 100 mA g^{-1} and outstanding rate capability. It also delivers the reversible capacity of 116.1 mAh g^{-1} with 71% capacity retention under a current density of 700 mA g^{-1} after 1000 cycles, which suggests an ultralong cycle life. In view of the abovementioned advantages of BNO-H, it can be said firmly that the material will be increasingly promising to a large extent in the near future.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jpowsour.2019.04.051>.

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